200mW, 2.4V - 51V Surface Mount Zener Diode

FEATURES

- AEC-Q101 qualified
- Zener voltage range selection: 2.4V to 51V
- V_Z tolerance selection of ± 2%
- Ideally suited for automated assembly processes
- Moisture sensitivity level: level 1, per J-STD-020
- RoHS Compliant
- Halogen-free

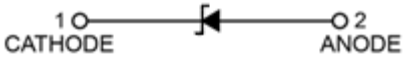
KEY PARAMETERS		
PARAMETER	**VALUE**	**UNIT**
P_D	200	mW
V_Z	2.4 - 51	V
$T_{J\ MAX}$	150	°C
V_F at I_F=10mA	0.9	V
Configuration	Single die	

APPLICATIONS

- General regulation functions

MECHANICAL DATA

- Case: SOD-523F
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Polarity: Indicated by cathode band
- Weight: 1.36mg (approximately)

PACKAGE: SOD-523F	PIN CONFIGURATION	CIRCUIT DIAGRAM
		

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)

PARAMETER	SYMBOL	VALUE	UNIT
Power dissipation(1)	P_D	200	mW
Forward voltage @ I_F=10mA	V_F	0.9	V
Junction temperature	T_J	-55 to +150	°C
Storage temperature	T_{STG}	-55 to +150	°C

Note:

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

THERMAL PERFORMANCE

PARAMETER	SYMBOL	TYP	UNIT
Junction-to-ambient thermal resistance(1)	$R_{\Theta JA}$	625	°C/W

Thermal Performance Note:

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

ELECTRICAL SPECIFICATIONS (T_A = 25°C unless otherwise noted)

PART NUMBER	DEVICE MARKING	ZENER VOLTAGE			TEST CURRENT	REGULAR IMPEDANCE		TEST CURRENT	LEAKAGE CURRENT		TEMPERATURE COEFFICIENT		TEST CURRENT
		V_Z @ I_{ZT}			I_{ZT}	Z_{ZT} @ I_{ZT}	Z_{ZK} @ I_{ZK}	I_{ZK}	I_R @ V_R		@ I_{ZT}		I_{ZT}
		V			mA	Ω	Ω	mA	μA	V	mV/°C		mA
		Min	Nom	Max		Max	Max		Max		Min	Max	
BZX585B2V4H	05	2.35	2.4	2.45	5	100	564	1	45	1	-3.5	0	5
BZX585B2V7H	15	2.65	2.7	2.75	5	100	564	1	18	1	-3.5	0	5
BZX585B3V0H	25	2.94	3.0	3.06	5	100	564	1	9.0	1	-3.5	0	5
BZX585B3V3H	35	3.23	3.3	3.37	5	95	564	1	4.5	1	-3.5	0	5
BZX585B3V6H	45	3.53	3.6	3.67	5	90	564	1	4.5	1	-3.5	0	5
BZX585B3V9H	55	3.82	3.9	3.98	5	90	564	1	2.7	1	-3.5	0	5
BZX585B4V3H	65	4.21	4.3	4.39	5	90	564	1	2.7	1	-3.5	0	5
BZX585B4V7H	75	4.61	4.7	4.79	5	80	470	1	2.7	2	-3.5	0	5
BZX585B5V1H	85	5.00	5.1	5.20	5	60	451	1	1.8	2	-2.7	1.2	5
BZX585B5V6H	95	5.49	5.6	5.71	5	40	376	1	0.9	2	-2.0	2.5	5
BZX585B6V2H	A5	6.08	6.2	6.32	5	10	141	1	2.7	4	0.4	3.7	5
BZX585B6V8H	B5	6.66	6.8	6.94	5	15	75	1	1.8	4	1.2	4.5	5
BZX585B7V5H	C5	7.35	7.5	7.65	5	15	75	1	0.9	5	2.5	5.3	5
BZX585B8V2H	D5	8.04	8.2	8.36	5	15	75	1	0.63	5	3.2	6.2	5
BZX585B9V1H	E5	8.92	9.1	9.28	5	15	94	1	0.45	6	3.8	7.0	5
BZX585B10H	F5	9.80	10	10.20	5	20	141	1	0.18	7	4.5	8.0	5
BZX585B11H	G5	10.78	11	11.22	5	20	141	1	0.09	8	5.4	9.0	5
BZX585B12H	H5	11.60	12	12.24	5	25	141	1	0.09	8	6.0	10.0	5
BZX585B13H	J5	12.74	13	13.26	5	30	160	1	0.09	8	7.0	11.0	5
BZX585B15H	K5	14.70	15	15.30	5	30	188	1	0.045	10.5	9.2	13.0	5
BZX585B16H	L5	15.68	16	16.32	5	40	188	1	0.045	11.2	10.4	14.0	5
BZX585B18H	M5	17.64	18	18.36	5	45	212	1	0.045	12.6	12.4	16.0	5
BZX585B20H	N5	19.60	20	20.40	5	55	212	1	0.045	14.0	14.4	18.0	5
BZX585B22H	P5	21.56	22	22.44	5	55	235	1	0.045	15.4	16.4	20.0	5
BZX585B24H	R5	23.52	24	24.48	5	70	235	1	0.045	16.8	18.4	22.0	5
BZX585B27H	S5	26.46	27	27.54	2	80	282	0.5	0.045	18.9	21.4	25.3	2
BZX585B30H	T5	29.40	30	30.60	2	80	282	0.5	0.045	21.0	24.4	29.4	2
BZX585B33H	U5	32.34	33	33.66	2	80	306	0.5	0.045	23.0	27.4	33.4	2
BZX585B36H	V5	35.28	36	36.72	2	90	329	0.5	0.045	25.2	30.4	37.4	2
BZX585B39H	X5	38.22	39	39.78	2	130	329	0.5	0.045	27.3	33.4	41.2	2
BZX585B43H	Y5	42.14	43	43.86	2	150	353	0.5	0.045	30.1	10.0	12.0	5
BZX585B47H	Z5	46.06	47	47.94	2	170	353	0.5	0.045	33.0	10.0	12.0	5
BZX585B51H	-5	49.98	51	52.02	2	180	376	0.5	0.045	35.7	10.0	12.0	5

ORDERING INFORMATION

ORDERING CODE(1)	PACKAGE	PACKING
BZX585BxxxH RSG	SOD-523F	8,000 / 7" Tape & Reel

Notes: 1. "xxx" defines voltage from 2.4V(BZX585B2V4H) to 51V(BZX585B51H)

CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.1 Power Dissipation Curve

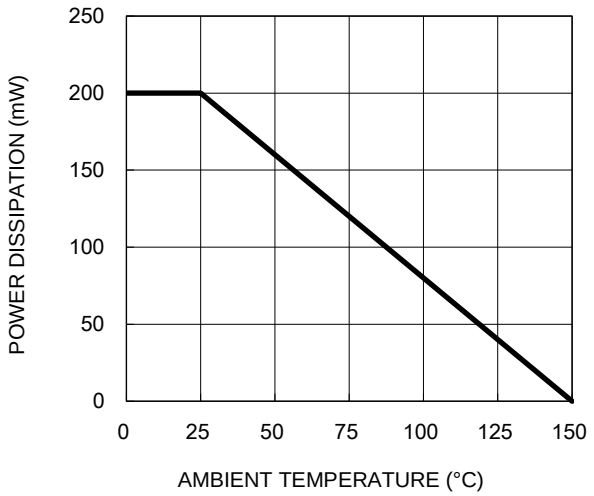


Fig.2 Zener Breakdown Characteristics

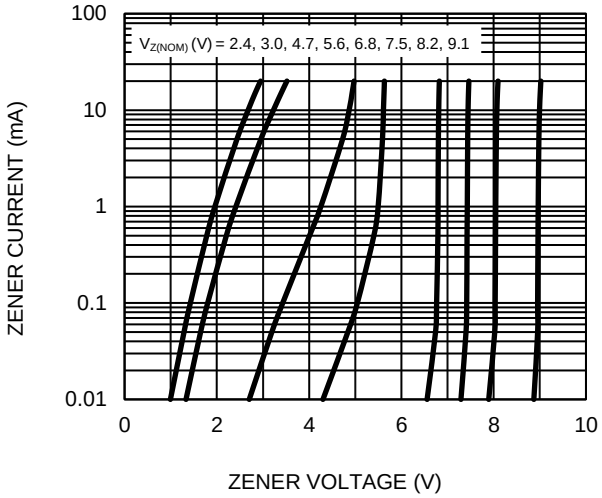


Fig.3 Zener Breakdown Characteristics

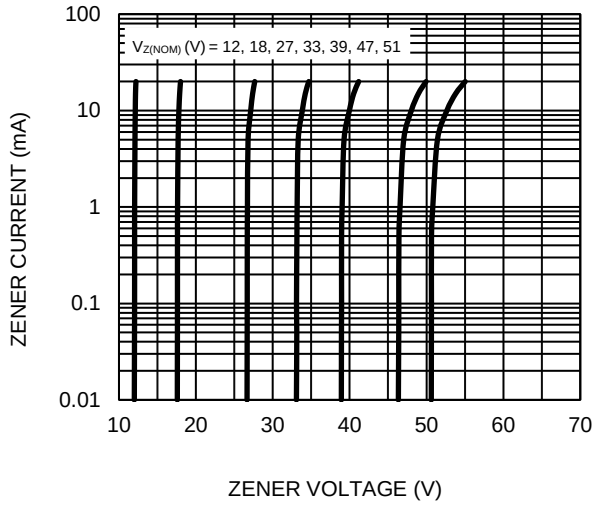


Fig.4 Typical Junction Capacitance

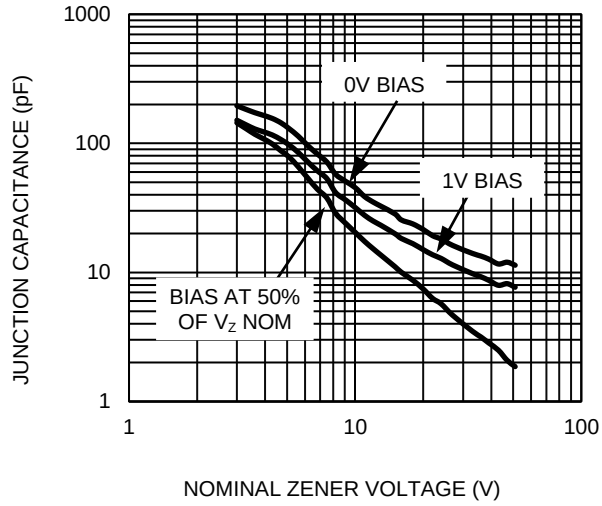
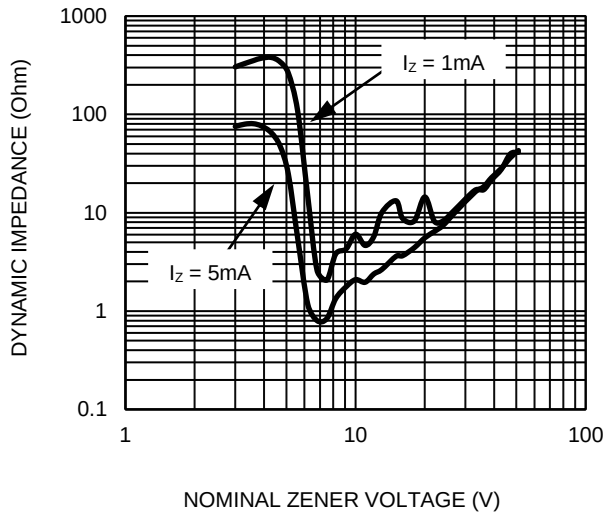
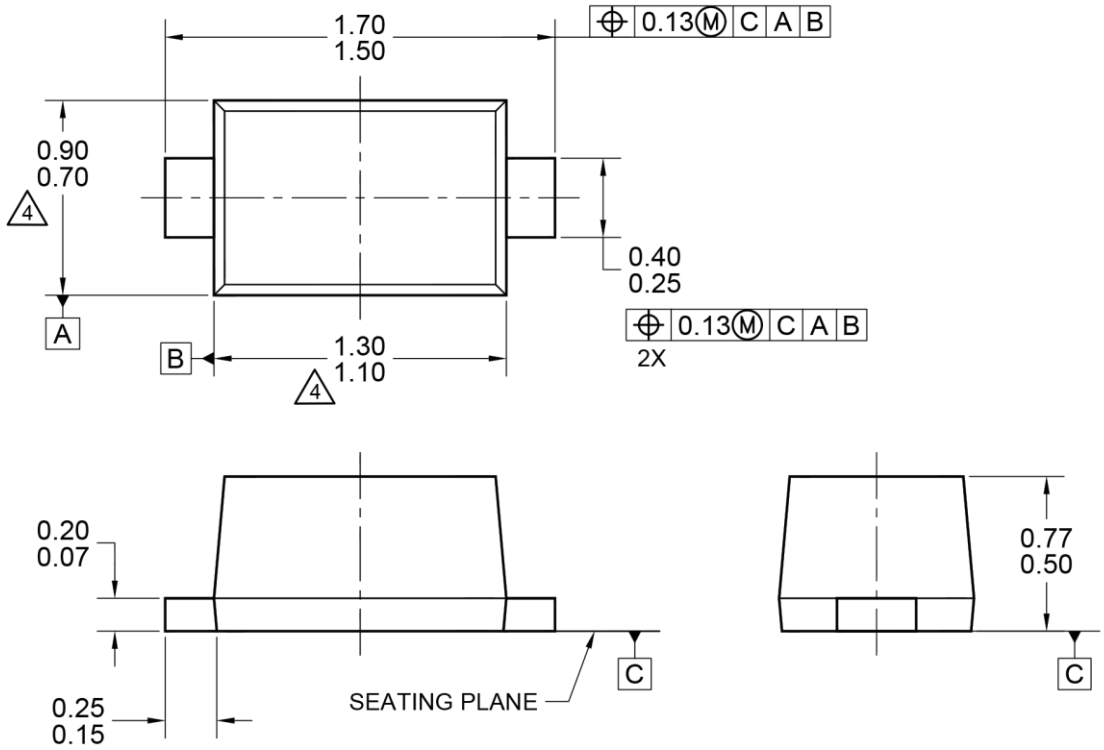


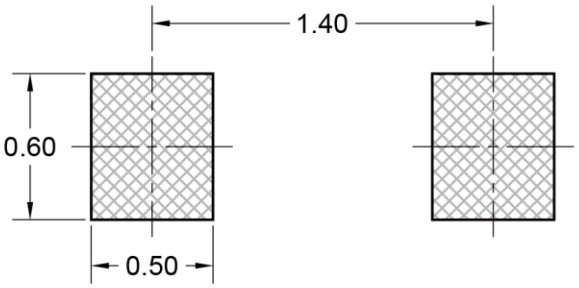
Fig.5 Effect of Zener Voltage on Impedance



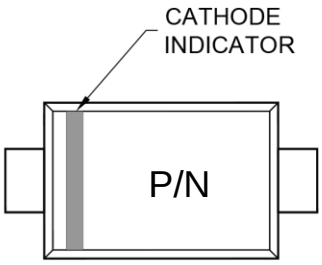
PACKAGE OUTLINE DIMENSION

SOD-523F





SUGGESTED PAD LAYOUT



MARKING DIAGRAM

P/N = Device marking

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. PACKAGE OUTLINE REFERENCE: EIAJ ED-7500A, SC-79.
4. MOLDED PLASTIC BODY LATERAL DIMENSIONS DO NOT INCLUDE MOLD FLASH.
5. DWG NO. REF: HQ2SD07-SOD523F-047 REV A.

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